CNC 3-Axis Router — United Control Box (RichAuto A11) — Design Spec and Commissioning Manual

Written in ASD-STE100 Simplified Technical English.

Rev	Date	Changes
0.1	2026-10-16	First draft. This design replaces the two-box NVEM/LinuxCNC design (CNC_Control_Panel_Spec.md). Keep the old spec for reference only. The new design has one enclosure, a RichAuto A11 standalone DSP controller and a 24 V DC operator panel.
0.2	2026-10-19	Text written again in ASD-STE100. There are no technical changes.
0.3	2026-10-19	§ 10 and wiring diagram sheets E1–E4 (A4 landscape) added.
0.4	2026-10-19	Terminal names changed to agree with the A11E board silkscreen. The board has no 0–10 V output. Spindle speed changed from analog to multi-speed (Y2–Y4). Wires 327–329 changed.
0.5	2026-10-19	Final manual. A11 manual, FC300 manual and DMA860H data added. Cautions C1–C13 (measure / confirm) added to § 5 and § 7 and to sheets E1–E4. Bench tests (§ 7.4) added.

1. Changes and reasons

Old design (2 boxes, NVEM)	New design (1 box, A11)	Reason
Box 1, Box 2 and Trunk A/B. Approx. 40 paired terminal blocks.	One enclosure. No trunks.	On 2026-10-16, most faults were at trunk wires and paired wires in the wrong position.
NVEM, Remora and LinuxCNC PC.	RichAuto A11 handheld and interface board. The G-code comes from a USB memory stick.	A PC, a real-time OS and HAL are not necessary.
220 V E-stop/ON/OFF chain directly to the contactor coil.	24 V E-stop/ON/OFF chain → 24 V relay K1. K1 supplies 220 V to the contactor coil.	There is no 220 V on the door.
220 V pilot lamps through a 6-ch relay module, MK3Pi relays and a 12 V buck.	24 V LED pilot lamps. The A11 outputs and the contactor aux contact control them directly.	This removes 4 parts and approx. 20 wires.
24 V PSU on the bus that the contactor switches.	24 V PSU on the unswitched side, with its own breaker.	The control chain must have power before the contactor closes. The A11 stays on and senses the E-stop.
0 V rail not connected to earth (floating).	0 V bonded to PE at one point (PELV).	If 220 V touches 0 V, the breaker trips. Before, the full 24 V side went to 230 V (fault found on 2026-10-16).
Two 48 V PSUs.	One 48 V/15 A PSU for X, Y and Z.	The 48 V/10.4 A unit is defective (rev 3.5).

The operator panel has only these items: E-stop, Power ON/OFF and three lamps. The A11 handheld has all the other controls: start, pause, stop, jog, zero and tool setting.

2. Parts

2.1 Parts that you have now (use again)

Part	Use
Main CB C25 2P	Incomer
EMI filter HD2AC-20A	Mains filter. Its rating is less than the C25 (refer to § 9 item 1).
Hyundai UMC65 22NS contactor, 220 VAC coil	Main power contactor (L and N poles). Aux 43-44 NO → yellow lamp.
CB C16A	VFD branch
CB C10A ×2	48 V PSU branch. Pump outlet branch (the old CB-PSU48Z).
CB C6A	24 V PSU branch (now unswitched)
VFD FC300-2.2G-S2-B4C and 100 Ω / 600 W braking resistor	Spindle drive. The parameters do not change (§ 7.3).
PSU 48 V / 15 A	Power for the three drives
PSU 24 V / 20 A	Power for the A11, sensors, panel, K1 and fan
3× Leadshine DMA860H	X, Y and Z drives. The DIP settings do not change: 1600 pulse/rev, 5.14 A peak.
Schneider XB2-BS542 E-stop (2× NC)	NC1 = 24 V control chain. NC2 = A11 X7.
Twin-head ON/OFF (ZB2-BE101C NO and ZB2-BE102C NC)	Power ON / OFF
HDC-HA-004 spindle connector, GX16-6 ×3 (motors), GX16-4 ×3 (limits), GX16-5 (tool setter)	Machine connectors. The pinouts do not change.
6× NPN NO proximity sensors, tool setter	No change
Schuko outlet, 24 V fan, earth bar, terminal blocks	Same as before

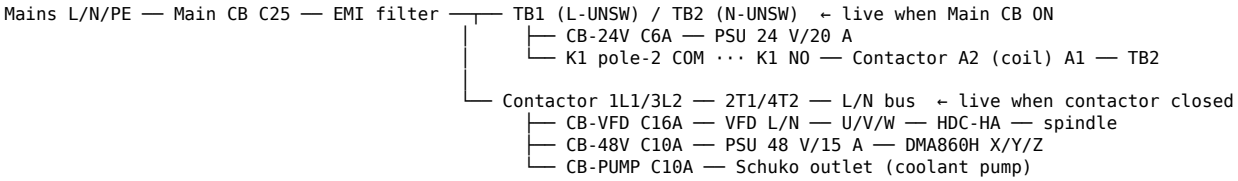
2.2 Parts to buy

Qty	Part	Notes
1	RichAuto A11E kit (handheld, interface board and 50-pin cable)	The board on hand has no 0–10 V analog output. The spindle uses multi-speed (S1–S3) — refer to § 5.2.
1	K1: relay with 24 VDC coil, 2 CO, 8 A 250 VAC minimum, with DIN socket and flyback diode module	Example: Finder 40.52.9.024, 95.05 socket and 99.02.9.024.99 diode/LED module.
3	22 mm 24 V DC LED pilot lamps: yellow, green, red	These replace the 220 V pilot lamps.
2	DIN fuse terminal 5×20 and 2 A fuses (and spare fuses)	F1 = control/lamps. F2 = A11/sensors.
—	End plates, partition plates, 2 jumper combs	Refer to § 6.
1	Enclosure, approx. 600 W × 800 H × 250 D mm, with steel backplate	Refer to § 6.
—	Wire duct 40×60, 0.5 mm² and 1.5 mm² wire, ferrules, wire markers	

2.3 Parts not used (do not put these parts in the new box)

NVEM. LinuxCNC PC (not necessary for operation). 6-ch relay module. 24→12 V buck. MK3Pi-1/-2 and sockets. 220 V pilot lamps. PSU 48 V/10.4 A (defective). All Trunk A/B cables and paired TBs. XB5 Cycle Start / Feed Hold / Stop (the A11 handheld does these functions). DB15 pendant bulkhead.

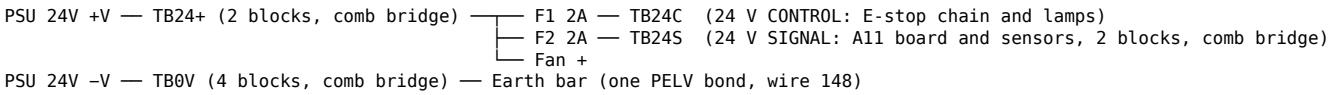
3. Power distribution (220 V)



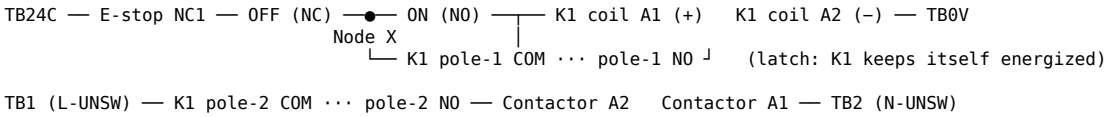
- When you push OFF or the E-stop, the VFD, the drives and the pump become de-energized. They all get power from the contactor bus.
- The A11 and the 24 V side stay energized. Thus the controller senses the E-stop and keeps its position data.
- The pump now has its own breaker. Before, the pump had no breaker.

4. 24 V control circuit

4.1 24 V distribution



4.2 Machine ON / OFF / E-stop (K1 self-latch)



- When you push ON, K1 energizes. Pole 1 latches K1. Pole 2 energizes the contactor coil. The bus becomes live.
- When you push OFF or the E-stop, the 24 V chain opens. K1 de-energizes. Then the contactor opens.
- If the mains power stops and then comes back, K1 stays de-energized. The machine does not start automatically.
- The contactor latch contact 13-14 is not used. K1 does the latch function. Do not connect wires to 13-14.

4.3 Lamps (24 V LED, + to 24 V, − switched)

Lamp	Indication	Wiring
Yellow	Power ON (contactor closed)	TB24C → contactor aux 43 · aux 44 → lamp + · lamp − → TB0V
Green	Machine in operation (A11 "Work" output)	TB24C → lamp + · lamp − → A11 Y6
Red	Alarm (A11 "Alarm" output)	TB24C → lamp + · lamp − → A11 Y5

The A11 Y outputs are low-level (sink) outputs. A 22 mm 24 V LED pilot lamp uses approx. 20 mA. The A11 manual does not give the Y-output current rating. Obey CAUTION C6 before you connect wires 224 and 226.

5. RichAuto A11 I/O map

The terminal names in this section agree with the silkscreen of the A11E board on hand (2026-10-19), the RichAuto A11 manual and the RichAuto A11 wiring diagram.

Block	Silkscreen terminals	Use
J2–J5: X_AXIS, Y_AXIS, Z_AXIS, C_AXIS	5V, PULSE, DIR, SHIELD	X, Y, Z: axis outputs (§ 5.3). C: not used. Do not connect SHIELD.

Block	Silkscreen terminals	Use
J7: Output signal	Y1 (S0), Y2 (S1), Y3 (S2), Y4 (S3), Y5, Y6, Y7, Y8, GND	Spindle and lamps (§ 5.2)
J8: Input signal	X1 (XH), X2 (YH), X3 (ZH), X4 (TS), X5, X6, X7, X8, 24V, GND, Shield	Sensors, alarms, E-stop (§ 5.1). Do not connect Shield.
J10: DC24V	24V, GND	Board power, 24 V / 3 A (wires 166, 167)

The board has no 0–10 V analog output. Thus the spindle speed uses the multi-speed outputs S1–S3 (8 speed grades).

5.1 Inputs (all are low-level active: input connected to GND = active)

A11	Function (A11 manual)	This build	Source	Level (Voltage Setup)
X1 (XH)	X origin	X home	X− sensor (X-LIM pin 1)	NO
X2 (YH)	Y origin	Y home	Y− sensor (Y-LIM pin 1)	NO
X3 (ZH)	Z origin	Z home	Z+ sensor (Z-LIM pin 2)	NO
X4 (TS)	Tool setting	Tool-setter touch	GX16-5 TS pin 1	NO
X5	Driver alarm (fixed)	Drive alarm X/Y/Z	DMA860H ALM	From bench test § 7.4 test 3
X6	Hard limit (fixed)	Far-end limits: X+, Y+, Z−	X-LIM pin 2, Y-LIM pin 2, Z-LIM pin 1 → TB-HL	NO
X7	E-stop (fixed)	E-stop NC2	XB2-BS542 second NC	NC
X8	Definable	Tool-setter over-travel	GX16-5 TS pin 3	Not in Voltage Setup. Do not connect (§ 9 item 7).
24V / GND	Sensor supply, DC 10–24 V	—	TB24S / TB0V (wires 168, 169)	—
Shield	—	Not used	—	—

You can connect NPN NO sensor outputs in parallel to one input. When one sensor operates, it pulls X6 to low level. Thus the three far-end limits use one input, X6.

⚠ **CAUTION C10 — MEASURE / CONFIRM.** Before you connect a sensor connector, measure pin 4 (BN) = +24 V and pin 3 (BU) = 0 V at the connector. Make sure that each sensor is NPN. A reversed supply can cause damage to the sensor. A voltage from the 48 V or 220 V side on an X input causes damage to the A11.

5.2 Outputs

A11	Function	This build
Y1 (S0)	Spindle ON / FWD	→ VFD X1 (forward run) · wire 325
Y2 (S1)	Speed bit 1	→ VFD X2 (multi-speed 1) · wire 327
Y3 (S2)	Speed bit 2	→ VFD X3 (multi-speed 2) · wire 328
Y4 (S3)	Speed bit 3	→ VFD X4 (multi-speed 3) · wire 329
Y5	Alarm LED	→ Red lamp − · wire 226
Y6	Work LED	→ Green lamp − · wire 224
Y7–Y8	Definable	Spare
GND	Output common	→ VFD COM (digital) · wire 326

The FC300 manual confirms that an FC300 input is active when it is connected to COM. Thus the A11 sink outputs Y1–Y4 connect directly to X1–X4. No relays are necessary.

⚠ **CAUTION C7 — MEASURE / CONFIRM.** Before you connect wires 325–329: with the VFD ON and the wires disconnected, measure X1, X2, X3 and X4 to VFD COM. Each must be a DC voltage of 24 V or less, X positive. Then measure VFD COM to TB0V (DC and AC ranges). It must be less than 1 V. If not, do not connect: a VFD COM with a mains voltage destroys the A11 output stage.

⚠ **CAUTION C6 — MEASURE / CONFIRM.** The A11 manual does not give the current rating of Y5 and Y6. Get the rating from the supplier. Measure the lamp current (§ 7.4 test 1). Connect the lamp only if its current is less than the rating. Do not connect a Y output to +24 V without a load.

5.3 Axis outputs (common anode)

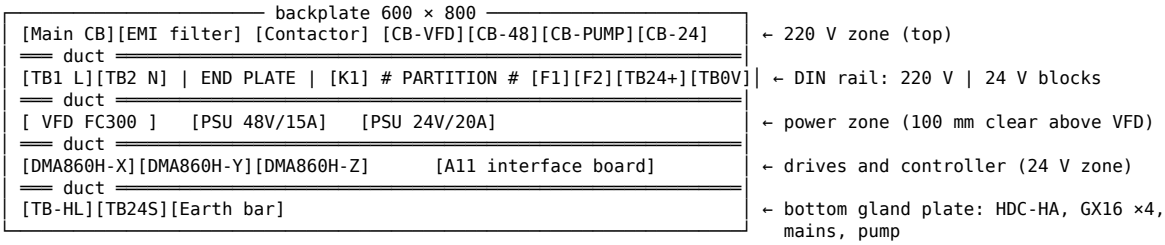
A11 axis block	DMA860H
5V	PUL+ and DIR+ (jumper at the drive)
PULSE	PUL−
DIR	DIR−
SHIELD	Do not connect. A shield is not necessary in one box (twisted wires, less than 1 m).

The A11 axis outputs supply 8 mA maximum. The DMA860H inputs need 7 mA minimum, 10 mA typical, 16 mA maximum, at 5 V. The margin is small.

> ⚠ **CAUTION C5 — MEASURE / CONFIRM.** Before you connect wires 301–312, measure A11 5V to GND = 5 V. Make sure that no 24 V or 48 V wire touches a 5V, PULSE or DIR terminal: 24 V on these terminals destroys the A11. After you connect, measure the current in wire 301, 305 and 309 (§ 7.4 test 2). It must be 7–8 mA.

6. Enclosure and layout

Size: approx. 600 W × 800 H × 250 D mm, with a full steel backplate. The 250 mm depth is sufficient for the VFD, the PSUs and the operators on the door.



Door: E-stop (top) · ON/OFF twin-head · yellow, green and red lamps · A11 handheld holder and cable gland.

Rules from the faults of 2026-10-16:

1. A feed-through block is one terminal. Its top screw and its bottom screw connect to the same conductor. Do not give them two different TB numbers.
2. Only TB24+, TB24S and TB0V have comb bridges. All other blocks are separate. Install an end plate where 220 V blocks and 24 V blocks are adjacent.
3. Put the § 8 number on each wire at the two ends. Read the marker in the correct orientation before you connect the wire.
4. Keep 220 V wires and 24 V wires in different ducts. If they must cross, they must cross at 90°.

7. Commissioning

7.0 Cautions (measure / confirm)

Each caution below stops a step that can cause damage to the A11, the drives, the VFD or the spindle. Do the measurement or the check before you do the step. If the result is not correct, stop and find the cause. Sheets E1–E4 show the same numbers with a ⚠ Cn mark.

No.	Before you…	Measure / confirm	Correct result	Risk if not
C1	Set the Main CB ON for the first time	Dead tests § 7.1, tests 1–3	220 V does not touch 24 V or PE. 0 V is bonded to PE.	220 V on the 24 V side destroys the A11, drives and sensors
C2	Connect A11 power (166–169)	G2 voltage and polarity at TB24S / TB0V	23–25 V DC, TB24S positive	Damage to the A11 (24 V / 3 A board)
C3	Connect drive power (150–155)	G1 output voltage	46–50 V DC (drive range 26–113 V DC)	Damage to the drives
C4	Connect or disconnect a motor connector	CB-48V is OFF. Drive LEDs are off.	No power on the drive	Damage to the drive output stage
C5	Connect axis signals (301–312)	A11 5V terminal. No 24 V or 48 V near signals. Loop current.	5 V. Current 7–8 mA.	24 V destroys the A11. Low current gives lost steps.
C6	Connect lamps to Y5 / Y6 (224, 226)	Lamp current. Y-output rating from supplier.	Lamp current less than the rating	Damage to the A11 output
C7	Connect VFD inputs (325–329)	X1–X4 to COM. VFD COM to TB0V.	≤ 24 V DC, X positive. COM to TB0V < 1 V.	Damage to the A11 output stage
C8	Run the spindle for the first time	FC300 parameters (§ 7.3)	P003 = P021 = 400.0, P022 = 100.0, P064 = 0, P092–P094 = 1/2/3, P095–P097 = 0	Overspeed of the spindle
C9	Connect ALM wires (315–320)	ALM polarity (§ 7.4 test 3)	Series or parallel and X5 level decided	False alarms or a hidden drive fault
C10	Connect sensor connectors	Pin 4 = +24 V, pin 3 = 0 V. Sensors are NPN.	Correct polarity	Damage to sensors or A11 inputs
C11	Push ON for the first time	K1 diode module polarity	Diode cathode on A1 (+)	Reverse diode shorts 24 V; F1 opens
C12	Set CB-VFD ON for the first time	Braking resistor terminals	R1 on P+ and PB only	Damage to the VFD
C13	Cut material with S commands	S-grade bit order (§ 7.4 test 4)	S1 = 000 … S8 = 111	Wrong spindle speed; tool damage

7.1 Tests with no power

Set the Main CB to OFF. Set the meter to Ω / continuity. Do these tests:

#	Test	Expected result
1	TB1 ↔ TB2	Open (or a high value through the bleed resistor of the EMI filter)
2	TB1 ↔ TB0V, TB1 ↔ TB24+, TB1 ↔ PE	Open. 220 V must not touch 24 V or earth.
3	TB0V ↔ PE	Beep (PELV bond, wire 148)
4	TB24C ↔ K1 A1, no button pushed	Open
5	Same as test 4, ON pushed and held	Beep
6	Same as test 4, ON held and E-stop pushed	Open
7	Same as test 4, ON held and OFF pushed	Open
8	TB1 ↔ contactor A2 (K1 not energized)	Open
9	Push the test lever of K1. TB1 ↔ contactor A2.	Beep
10	E-stop NC2: A11 X7 ↔ TB0V	Beep when the E-stop is released. Open when it is pushed.
11	PSU 48V +V ↔ −V	Not 0 Ω
12	TB24+ ↔ TB0V	Not 0 Ω
13	5V, PULSE, DIR of each axis block ↔ TB24+ and ↔ PSU 48V +V	Open

⚠ **CAUTION C1 — MEASURE / CONFIRM.** Do not set the Main CB ON until tests 1, 2, 3, 12 and 13 are correct. 220 V or 24 V on the 5 V side destroys the A11 and the drives.

⚠ **CAUTION C12 — MEASURE / CONFIRM.** Make sure that R1 connects to VFD terminals P+ and PB only. Read the terminal labels on the VFD.

7.2 Power-up in steps

WARNING: WHEN THE MAIN CB IS ON, THERE IS 220 V ON THE BACKPLATE. DO NOT TOUCH THE TERMINALS DURING THIS PROCEDURE. 220 V CAN KILL YOU.

Stage 1 — 24 V side and K1

1. Make sure that § 7.1 is complete.
2. Disconnect contactor coil wire 131. Set QF1, QF2 and QF3 to OFF. Disconnect wires 166 and 168 at the A11.
3. Set the Main CB to ON.

⚠ **CAUTION C2 — MEASURE / CONFIRM.** Measure TB24S to TB0V. The voltage must be 23–25 V DC, with TB24S positive. If not, do not connect the A11.

4. Set the Main CB to OFF. Connect wires 166 and 168. Set the Main CB to ON. Make sure that the A11 starts.

⚠ **CAUTION C11 — MEASURE / CONFIRM.** Before you push ON, make sure that the K1 diode module has its cathode (+) on A1. A reversed diode shorts the 24 V control supply.

5. Push ON. Make sure that K1 energizes and stays energized.
6. Push OFF. Make sure that K1 de-energizes.
7. Push ON again. Push the E-stop. Make sure that K1 de-energizes and stays de-energized.

Stage 2 — Contactor

8. Release the E-stop. Set the Main CB to OFF. Connect wire 131. Set the Main CB to ON.
9. Push ON. Make sure that the contactor closes and the yellow lamp comes on.
10. Push OFF. Make sure that the contactor opens. Do this test again with the E-stop.

Stage 3 — Drives and axes

⚠ **CAUTION C3 — MEASURE / CONFIRM.** Disconnect wires 150–155 at the drives. Set QF2 to ON. Measure the G1 output. It must be 46–50 V DC. If not, adjust G1. Set QF2 to OFF.

⚠ **CAUTION C4 — MEASURE / CONFIRM.** Connect all motor connectors before you set QF2 to ON. Do not connect or disconnect a motor connector when QF2 is ON.

11. Connect wires 150–155. Set QF2 to ON. Make sure that the drive LEDs are green.

⚠ **CAUTION C5 — MEASURE / CONFIRM.** Measure the axis current: do § 7.4 test 2 before you use high feed rates.

12. Use the handheld to jog each axis at low speed. Make sure that each axis moves in the correct direction.
13. Do § 7.4 test 2 (pulse current and lost steps).

> ⚠ **CAUTION C9 — MEASURE / CONFIRM.** Do § 7.4 test 3 (ALM polarity) before you connect wires 315–320 to X5.

14. Connect wires 315–320 as test 3 shows. Set X5 in MACHINE SETUP → Voltage Setup.

Stage 4 — Spindle

15. Disconnect wires 325–329 at the VFD. Set QF1 to ON. Set all FC300 parameters in § 7.3.

> ⚠ **CAUTION C8 — MEASURE / CONFIRM.** Read back P003, P021, P022, P064 and P092–P097 before the first run. A wrong P003 or P021 can overspeed the spindle.

> ⚠ **CAUTION C7 — MEASURE / CONFIRM.** Measure X1–X4 to VFD COM (24 V DC or less, X positive). Measure VFD COM to TB0V (less than 1 V DC and AC). If not correct, do not connect.

16. Set QF1 to OFF. Connect wires 325–329. Set QF1 to ON.

> ⚠ **CAUTION C13 — MEASURE / CONFIRM.** Do § 7.4 test 4 (S-grade bit order) before you cut material.

17. Do § 7.4 test 4.
18. Do a test of M3 at each speed grade S1–S8. For each grade, make sure that the VFD shows the frequency in the § 7.3 table.
19. Make sure that the E-stop stops the spindle.

Stage 5 — Lamps, sensors and limits

> ⚠ **CAUTION C6 — MEASURE / CONFIRM.** Do § 7.4 test 1 (lamp current) before you connect wires 224 and 226.

20. Connect wires 224 and 226 only if test 1 is correct.

> ⚠ **CAUTION C10 — MEASURE / CONFIRM.** Before you connect each GX16-4 and GX16-5 connector, measure pin 4 = +24 V and pin 3 = 0 V at the connector.

21. Connect the sensor connectors. In SYSTEM SETUP → Input Port, make sure that each sensor changes its input when you operate it.
22. Home the three axes.
23. Operate each hard-limit sensor by hand. Make sure that the A11 stops and shows an alarm, and that the red lamp comes on.

7.3 Settings

A11 settings:

Setting	Menu (A11 manual)	Value
Pulse equivalent X/Y/Z	MACHINE SETUP (refer to the A11 manual)	320 pulse/mm (0.003125 mm/pulse) = 1600 pulse/rev ÷ 5 mm (SFU2005)
Home direction	MACHINE SETUP	X −, Y −, Z +
Input levels	MACHINE SETUP → Voltage Setup (↓ = NO, ↑ = NC)	X1–X4, X6 = NO. X7 = NC. X5 = result of § 7.4 test 3.
Unused inputs	MACHINE SETUP → Input Confi	Disable X8 (not connected)
Input check	SYSTEM SETUP → Input Port	Each input changes when its sensor operates
Spindle	MACHINE SETUP → Spindle Setup → Spindle state	3-line, 8-state. S1–S8 rpm as the table below. Start / stop delay as necessary.
Table size / soft limits	MACHINE SETUP	Real travel of the machine

Spindle speed table (2-pole spindle, 400 Hz = 24000 rpm). The FC300 table is from the FC300 manual (attached table 1). The A11 bit order is not in the RichAuto documents. Confirm it with § 7.4 test 4.

A11 grade	Y4 Y3 Y2 (expected)	VFD X4 X3 X2	FC300 param	Frequency	Speed
S1	0 0 0	off off off	P030	100.0 Hz	6000 rpm
S2	0 0 1	off off on	P031	133.3 Hz	8000 rpm
S3	0 1 0	off on off	P032	166.7 Hz	10000 rpm
S4	0 1 1	off on on	P033	200.0 Hz	12000 rpm
S5	1 0 0	on off off	P034	250.0 Hz	15000 rpm
S6	1 0 1	on off on	P035	300.0 Hz	18000 rpm
S7	1 1 0	on on off	P036	350.0 Hz	21000 rpm
S8	1 1 1	on on on	P037	400.0 Hz	24000 rpm

1 = output active (pulled to GND) = VFD input on.

FC300 parameters:

Param	Value	Function (FC300 manual)
P001	8.0	From the old spec (no change)
P002	220	Motor rated voltage
P003	400.0	Motor rated frequency
P021	400.0	Maximum frequency
P022	100.0	Minimum frequency (6000 rpm)
P030–P037	Speed table	First to eighth frequency (multi-speed)
P058	8–10 kHz approx.	From the old spec (no change)
P064	0	Frequency source: panel digital (was 1 = AVI)
P065	1	Run command: external terminals (X1 + COM = forward run)
P066	0	Steering control (direction). Change to 1 if the spindle turns the wrong way.
P067	0	Reverse run not permitted
P091	8	X1 function: forward
P092 / P093 / P094	1 / 2 / 3	X2 / X3 / X4: first / second / third multi-speed
P095 / P096 / P097	0 / 0 / 0	X5 / X6 / X7: no effect (factory setting is multi-speed)

Do not use the ▲ / ▼ keys on the VFD panel. They change P030 (speed grade S1), and the VFD keeps the new value after a power cut.

7.4 Bench tests

Test 1 — Lamp current (C6).

1. Connect a lamp + to TB24C. Connect lamp − through a meter (mA range) to TB0V.
2. Read the current. Write it in the build record.
3. Compare it with the Y-output rating from the supplier. Connect the lamp to Y5 or Y6 only if the current is less than the rating.

Test 2 — Pulse current and lost steps (C5).

1. Disconnect wire 301 at PUL+. Connect a meter (mA range) between A11 X_AXIS 5V and PUL+.
2. Jog X at high speed. The current must be 7–8 mA. Do the same test for Y (305) and Z (309).
3. Connect the wires again. Put a mark at zero on each axis.
4. Run a long G-code program at full feed rate. Then go back to zero.
5. Make sure that each axis stops on its mark. An offset shows lost pulses. If there is an offset, install a 5 V line driver between the A11 and the drives.

Test 3 — DMA860H ALM polarity (C9).

1. Connect a 2.2 kΩ resistor from TB24C to ALM+ of one drive. Connect ALM− to TB0V.
2. Set QF2 to ON. Measure ALM+ to TB0V with no fault.
3. Set QF2 to OFF. Disconnect the motor connector of that drive. Set QF2 to ON. The drive shows a fault. Measure again.
4. Low voltage = the opto conducts. If it conducts in normal operation: connect the three ALM outputs in series and set X5 = NC. If it conducts only on a fault: connect them in parallel and set X5 = NO.
5. Set QF2 to OFF. Remove the resistor. Connect the motor connector again.

Test 4 — S-grade bit order (C13).

1. Disconnect wires 327, 328 and 329 at the VFD, or set QF1 to OFF.
2. Connect a meter (DC V) between Y2 and GND, then Y3 and GND, then Y4 and GND.
3. Start the spindle on the handheld and select S1 to S8 one at a time. Write the Y4 Y3 Y2 pattern for each grade.
4. Compare the result with the § 7.3 table. If the order is different, change only P030–P037. The wiring does not change.
5. Run G-code with S6000, S13000 and S24000. Write which grade the A11 selects for each value.

8. Wire schedule

Wire numbers: 1xx = 220 V power and PE. 2xx = 24 V control and lamps. 3xx = signals.

Wire sizes: 220 V = 1.5 mm² (2.5 mm² for mains, VFD and bus). 24 V and signals = 0.5 mm². PE = green/yellow (G/Y).

8.1 220 V power

No.	From	To	Function	Spec
101	Mains L (gland)	Main CB L in	Mains L	2.5 brown

No.	From	To	Function	Spec
102	Mains N (gland)	Main CB N in	Mains N	2.5 blue
103	Main CB L out	EMI L-IN		2.5 brown
104	Main CB N out	EMI N-IN		2.5 blue
105	EMI L-OUT	TB1 (L-UNSW)	Unswitched L	2.5 brown
106	EMI N-OUT	TB2 (N-UNSW)	Unswitched N	2.5 blue
107	TB1	Contactor 1L1	Contactor L supply	2.5 brown
108	TB2	Contactor 3L2	Contactor N supply	2.5 blue
109	Contactor 2T1	L bus	Switched L	2.5 brown
110	Contactor 4T2	N bus	Switched N	2.5 blue
111	TB1	CB-24V (C6A) in	24 V PSU L (unswitched)	1.5 brown
112	CB-24V out	PSU 24V L		1.5 brown
113	TB2	PSU 24V N		1.5 blue
114	L bus	CB-VFD (C16A) in		2.5 brown
115	CB-VFD out	VFD L		2.5 brown
116	N bus	VFD N		2.5 blue
117	L bus	CB-48V (C10A) in		1.5 brown
118	CB-48V out	PSU 48V L		1.5 brown
119	N bus	PSU 48V N		1.5 blue
120	L bus	CB-PUMP (C10A) in		1.5 brown
121	CB-PUMP out	Schuko L	Pump	1.5 brown
122	N bus	Schuko N	Pump	1.5 blue
123	VFD U	HDC-HA pin 1	Spindle U	1.5 shielded
124	VFD V	HDC-HA pin 2	Spindle V	1.5 shielded
125	VFD W	HDC-HA pin 3	Spindle W	1.5 shielded
126	VFD PE	HDC-HA PE	Spindle PE	G/Y
127	HDC-HA pin 4	VFD PE	Spindle shield (connect at VFD end only)	—
128	VFD P+	Braking resistor T1		1.5
129	VFD PB	Braking resistor T2		1.5
130	TB1	K1 pole-2 COM	Coil supply L	1.5 brown
131	K1 pole-2 NO	Contactor A2	Switched coil L	1.5 brown
132	TB2	Contactor A1	Coil N	1.5 blue

8.2 PE

No.	From	To	Function	Spec
140	Mains PE (gland)	Earth bar		4 G/Y
141	Backplate bond	Earth bar		4 G/Y
142	Door bond strap	Earth bar		2.5 G/Y
143	EMI filter ground lug	Earth bar		4 G/Y
144	VFD PE	Earth bar		2.5 G/Y
145	PSU 48V PE	Earth bar		1.5 G/Y
146	PSU 24V PE	Earth bar		1.5 G/Y
147	DIN-rail earth TB	Earth bar		G/Y
148	TB0V	Earth bar	PELV bond: 0 V to PE. This is the only 0 V to PE bond.	1.5 G/Y

8.3 DC power

No.	From	To	Function	Spec
150	PSU 48V +V (screw 1)	DMA860H-X AC 1	48 V	1.5 red
151	PSU 48V −V (screw 1)	DMA860H-X AC 2		1.5 black
152	PSU 48V +V (screw 2)	DMA860H-Y AC 1		1.5 red
153	PSU 48V −V (screw 2)	DMA860H-Y AC 2		1.5 black
154	PSU 48V +V (screw 3)	DMA860H-Z AC 1		1.5 red
155	PSU 48V −V (screw 3)	DMA860H-Z AC 2		1.5 black
160	PSU 24V +V	TB24+	24 V	1.5 red
161	PSU 24V −V	TB0V	0 V	1.5 black
162	TB24+	F1 in	Control fuse	0.5 red

No.	From	To	Function	Spec
163	F1 out	TB24C	24 V control	0.5 red
164	TB24+	F2 in	Signal fuse	0.5 red
165	F2 out	TB24S	24 V signal	0.5 red
166	TB24S	A11 board power +		0.5 red
167	TB0V	A11 board power −		0.5 black
168	TB24S	A11 input 24V+	Sensor/input supply	0.5 red
169	TB0V	A11 input GND		0.5 black
170	TB24+	Fan +		0.5 red
171	TB0V	Fan −		0.5 black

If the 48 V PSU has only two +V/−V screw pairs, connect 154/155 to a small 48 V block pair with a comb bridge. Do not put two wires under one screw.

8.4 24 V control and lamps

No.	From	To	Function
201	TB24C	E-stop NC1 in	Chain
202	E-stop NC1 out	OFF (NC) in	Chain
203	OFF out (Node X)	ON (NO) in	Chain
204	ON out	K1 coil A1	Start
205	Node X (OFF out)	K1 pole-1 COM	Latch
206	K1 pole-1 NO	K1 coil A1	Latch
207	K1 coil A2	TB0V	Coil return
210	E-stop NC2	A11 X7	E-stop signal to controller
211	E-stop NC2 (other screw)	TB0V	
220	TB24C	Contactor aux 43	Yellow lamp supply
221	Contactor aux 44	Yellow lamp +	
222	Yellow lamp −	TB0V	
223	TB24C	Green lamp +	
224	Green lamp −	A11 Y6	Work
225	TB24C	Red lamp +	
226	Red lamp −	A11 Y5	Alarm

Put the door wires (201–206, 210–211, 220–226) in one bundle. Make a service loop at the hinge.

8.5 Signals

No.	From	To	Function
301	A11 X-axis 5V	DMA860H-X PUL+	
302	A11 X-axis PULSE	DMA860H-X PUL−	
303	A11 X-axis DIR	DMA860H-X DIR−	
304	DMA860H-X PUL+	DMA860H-X DIR+	Jumper
305	A11 Y-axis 5V	DMA860H-Y PUL+	
306	A11 Y-axis PULSE	DMA860H-Y PUL−	
307	A11 Y-axis DIR	DMA860H-Y DIR−	
308	DMA860H-Y PUL+	DMA860H-Y DIR+	Jumper
309	A11 Z-axis 5V	DMA860H-Z PUL+	
310	A11 Z-axis PULSE	DMA860H-Z PUL−	
311	A11 Z-axis DIR	DMA860H-Z DIR−	
312	DMA860H-Z PUL+	DMA860H-Z DIR+	Jumper
315	DMA860H-X ALM+	A11 X5	Drive alarm (wire-OR)
316	DMA860H-X ALM−	TB0V	
317	DMA860H-Y ALM+	A11 X5	
318	DMA860H-Y ALM−	TB0V	
319	DMA860H-Z ALM+	A11 X5	
320	DMA860H-Z ALM−	TB0V	
325	A11 Y1	VFD X1	Spindle run
326	A11 output GND	VFD COM (digital)	Output common
327	A11 Y2 (S1)	VFD X2	Multi-speed bit 1
328	A11 Y3 (S2)	VFD X3	Multi-speed bit 2
329	A11 Y4 (S3)	VFD X4	Multi-speed bit 3

No.	From	To	Function
330	X-LIM pin 1	A11 X1	X home (X−)
331	Y-LIM pin 1	A11 X2	Y home (Y−)
332	Z-LIM pin 2	A11 X3	Z home (Z+)
333	X-LIM pin 2	TB-HL	X+ limit
334	Y-LIM pin 2	TB-HL	Y+ limit
335	Z-LIM pin 1	TB-HL	Z− limit
336	TB-HL	A11 X6	Hard limit (wire-OR)
337	TB0V	X-LIM pin 3	Sensor 0 V (BU)
338	TB0V	Y-LIM pin 3	
339	TB0V	Z-LIM pin 3	
340	TB24S	X-LIM pin 4	Sensor +24 V (BN)
341	TB24S	Y-LIM pin 4	
342	TB24S	Z-LIM pin 4	
345	TS pin 1 (red)	A11 X4	Tool-setter touch
346	TS pin 2 (black)	TB0V	
347	TS pin 3 (blue)	A11 X8	Tool-setter over-travel (NC). Do not connect (§ 9 item 7).
348	TS pin 4 (yellow)	TB0V	

The motor cores (DMA860H A+/A−/B+/B− → GX16-6 pins 1–4, ×3) are the same as in the old spec.

Total: approx. 117 numbered wires and 12 motor cores. The old design had 183 wires. There is no trunk and there are no paired terminal blocks.

9. Open items and settled items

9.1 Open items (close these before final assembly)

1. **EMI filter 20 A and Main CB C25.** Install a filter of 25 A or more, or replace the Main CB with a C20.
2. **Contactor coil inrush and K1 contact rating.** The UMC65 AC coil inrush is usually some hundred VA, much less than the 8 A / 250 VAC K1 contact. Make sure with the UMC65 datasheet.
3. **A11 Y-output current rating.** Not in the A11 manual or wiring diagram. Get it from the supplier (C6, § 7.4 test 1).
4. **DMA860H pulse current.** The A11 supplies 8 mA maximum. The drive needs 7 mA minimum, 10 mA typical. Do § 7.4 test 2 (C5).
5. **DMA860H ALM polarity and rating.** Not in the Leadshine data. Do § 7.4 test 3 (C9).
6. **A11 S-grade bit order.** Not in the RichAuto documents. Do § 7.4 test 4 (C13).
7. **X8 tool-setter over-travel.** The A11 manual has no setting for X8 function or level. Do not connect wire 347. Disable X8 in Input Confi.
8. **K1 contact numbers.** Sheets E1/E2 use Finder 40.52 numbers (11-14 latch, 21-24 KM1 coil). Compare them with the relay that you buy.
9. **48 V PSU output screws.** If the PSU has only two +V/−V pairs, connect 154/155 to a comb-bridged 48 V block pair.

9.2 Settled items (2026-10-19)

- A11E terminal names: silkscreen examined (§ 5).
- No analog output on the A11E board on hand: spindle speed uses multi-speed S1–S3 (§ 5.2).
- X5 = driver alarm, X6 = hard limit, X7 = E-stop: fixed functions (A11 manual). X7 = NC is set in Voltage Setup.
- FC300 inputs are active when connected to COM (sink). Terminal names X1–X8, COM confirmed. Multi-speed parameters P030–P037, P064, P091–P097 confirmed (FC300 manual).
- DMA860H supply 26–113 V DC: the 48 V supply is correct. Signal level 5 V, pulse width 2.5 µs minimum, 200 kHz maximum (Leadshine data).

9.3 References

- RichAuto DSP A11 manual and DSP A11 wiring diagram (richautocontroller.com).
- BEST FC300 inverter manual (English).
- Leadshine DMA860H product data (leadshine.com).

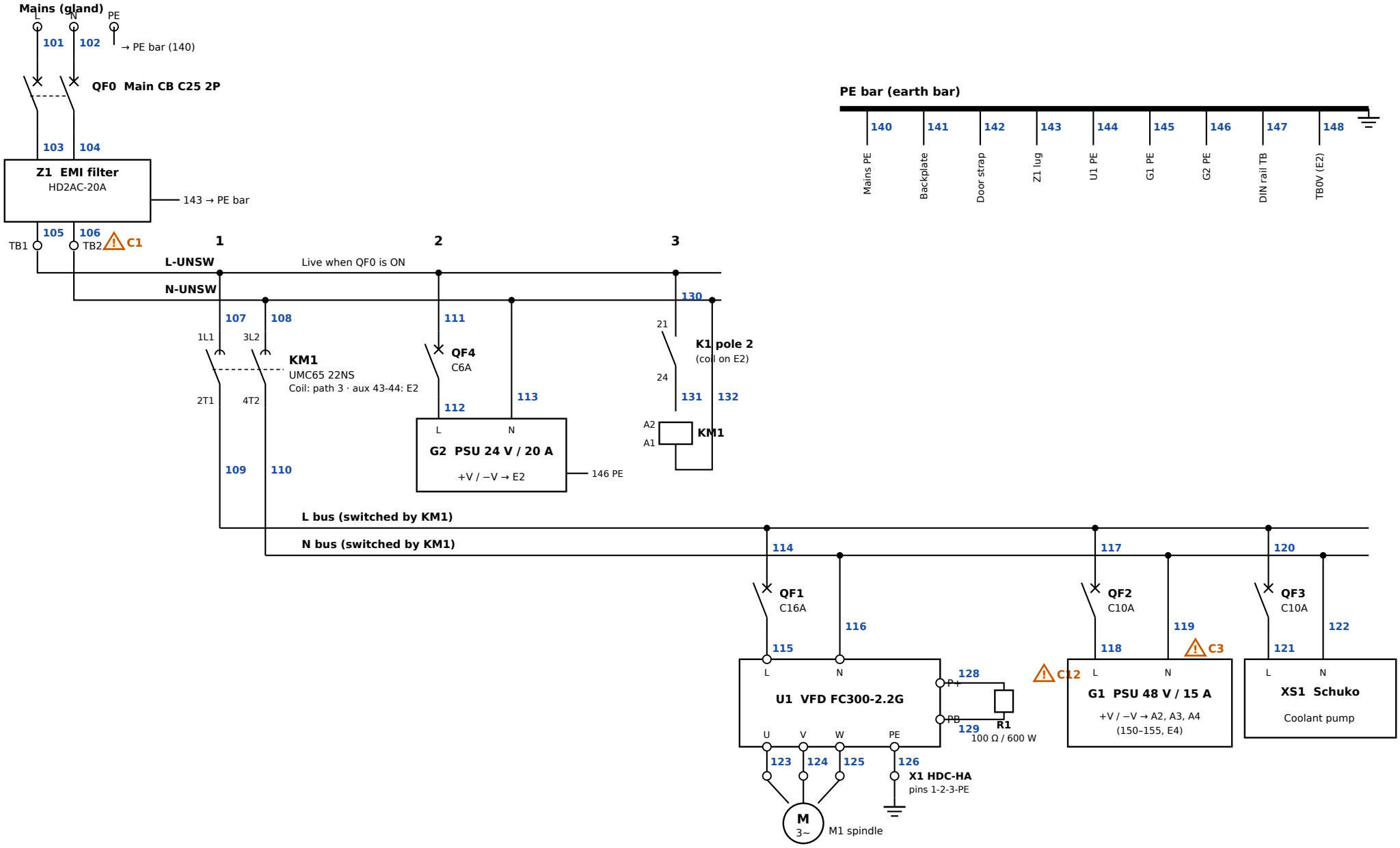
10. Wiring diagrams

Sheets E1 to E4 follow this page. They are A4 landscape. They use IEC 60617 symbols. The wire numbers on the sheets agree with § 8. If a sheet and § 8 do not agree, stop and correct the error before you connect wires.

Sheet	Title	Contents
E1	220 V power distribution and contactor	Mains, QF0, Z1, TB1/TB2, KM1 main contacts, L/N bus, QF1–QF4, U1, G1, G2, XS1, PE bar
E2	24 V control — K1 self-hold, E-stop, lamps	G2 outputs, F1/F2, S0/S1/S2 chain, K1 latch, H1–H3 lamps, PELV bond
E3	RichAuto A11 inputs	Limit and home sensors, TB-HL, tool setter, drive alarms, E-stop NC2
E4	RichAuto A11 outputs	Axis outputs to A2–A4, motor connectors, VFD run and speed, lamp outputs, board power

Device tags on the sheets:

Tag	Device	Tag	Device
QF0	Main CB C25 2P	S0	E-stop XB2-BS542 (NC1, NC2)
QF1	CB-VFD C16A	S1 / S2	OFF (NC) / ON (NO)
QF2	CB-48V C10A	H1 / H2 / H3	Lamps yellow / green / red
QF3	CB-PUMP C10A	A1	RichAuto A11E interface board
QF4	CB-24V C6A	A2 / A3 / A4	DMA860H X / Y / Z
Z1	EMI filter HD2AC-20A	U1	VFD FC300-2.2G
KM1	Contactor UMC65 22NS	R1	Braking resistor 100 Ω / 600 W
K1	Relay 24 VDC, 2 CO (Finder 40.52)	M1 / M2	Spindle / fan
F1 / F2	Fuses 2 A (control / signal)	X1	Spindle connector HDC-HA-004
G1 / G2	PSU 48 V / PSU 24 V	XS1	Schuko outlet (pump)
X-LIM / Y-LIM / Z-LIM	Limit connectors GX16-4	XT / XM-X, Y, Z	Tool setter GX16-5 / motor GX16-6



NOTES ⚠ **Cn** = CAUTION Cn: measure / confirm before you connect. Refer to spec § 7.0.

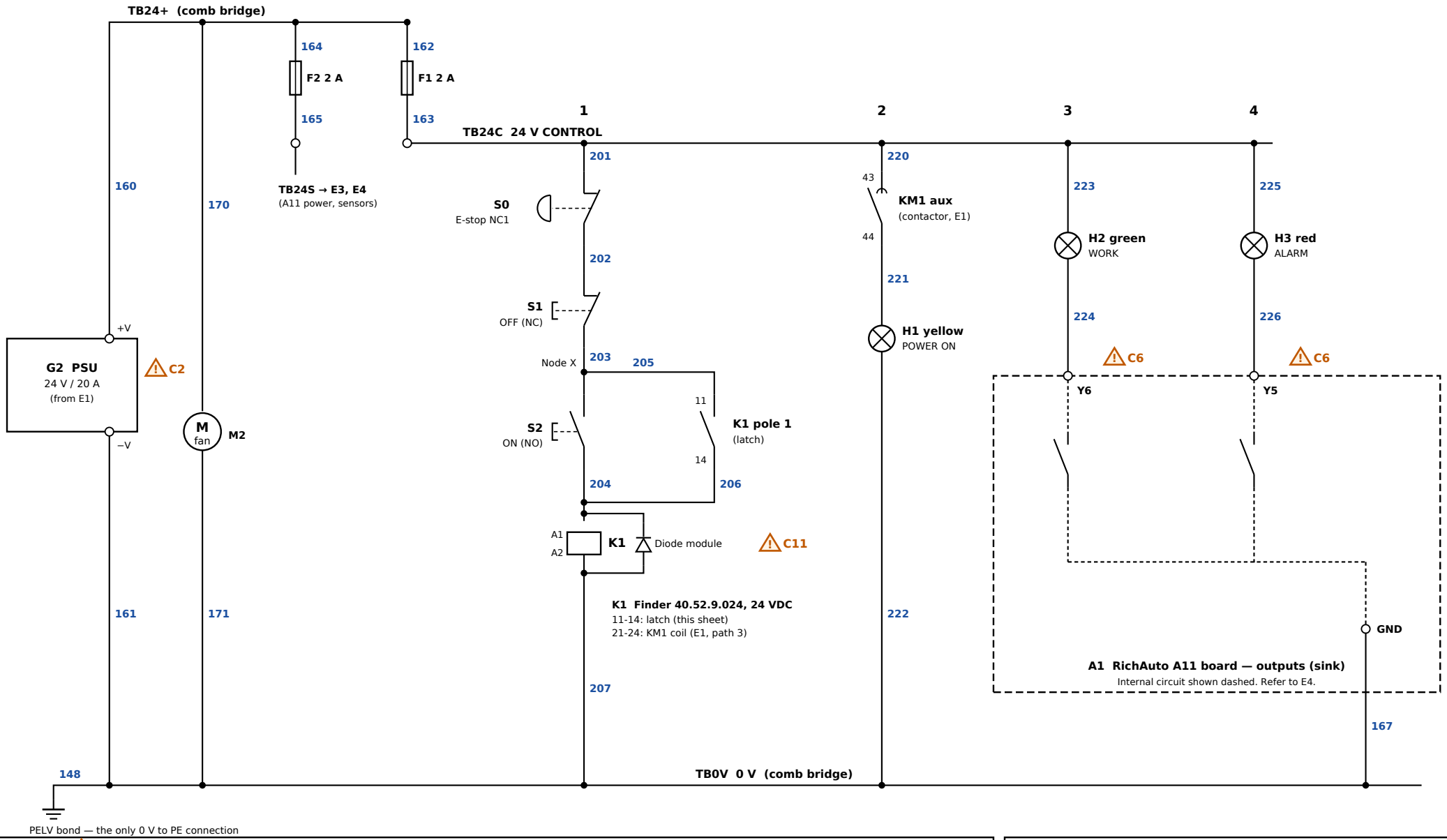
1. QF0, Z1, TB1/TB2, QF4, G2 and the KM1 coil circuit are live when QF0 is ON.
2. The L/N bus is live only when KM1 is closed. KM1 closes only when K1 is energized (E2).
3. When KM1 opens, U1 (VFD), G1 (48 V drives) and XS1 (pump) become de-energized.
4. Connect the spindle cable shield (HDC-HA pin 4, wire 127) to U1 PE. Connect it at the VFD end only.
5. Keep 100 mm clear above U1. Do not connect wires to KM1 13-14 (K1 does the latch function).
6. Wire sizes: 2.5 mm² for mains, bus and VFD. 1.5 mm² for other 220 V wires. PE = green/yellow.

CNC 3-Axis Router — United Control Box (RichAuto A11)

220 V power distribution and contactor

Sheet E1 of E4 · Rev 0.5 · 2026-10-19

IEC 60617 symbols · Text: ASD-STE100 · Wire nos. = spec § 8



NOTES ⚠ **Cn** = CAUTION Cn: measure / confirm before you connect. Refer to spec § 7.0.

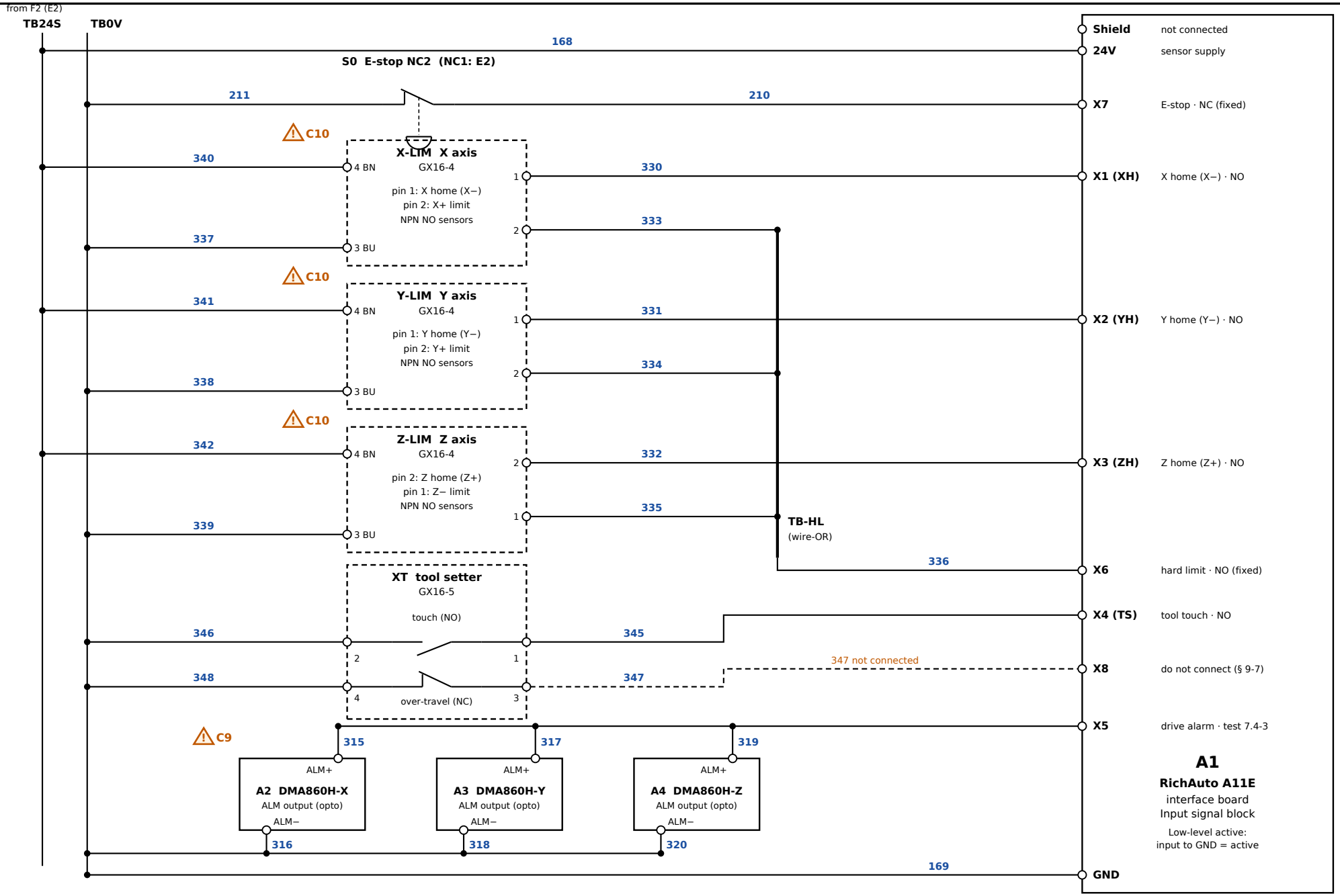
1. Push S2 (ON). K1 energizes. K1 11-14 closes and keeps K1 energized. K1 21-24 energizes the KM1 coil (E1).
2. Push S1 (OFF) or S0 (E-stop). The chain opens. K1 de-energizes. Then KM1 opens.
3. After a power failure, K1 stays de-energized. The machine does not start automatically. Push S2 to start.
4. S0 has a second contact (NC2). NC2 gives the E-stop signal to A11 input X7 (E3).
5. G2 is on the unswitched side (E1). Thus the A11 stays energized and senses the E-stop.
6. Connect 0 V to PE at one point only (wire 148). Do not connect 0 V to PE at other points.

CNC 3-Axis Router — United Control Box (RichAuto A11)

24 V control — K1 self-hold, E-stop, lamps

Sheet E2 of E4 · Rev 0.5 · 2026-10-19

IEC 60617 symbols · Text: ASD-STE100 · Wire nos. = spec § 8



NOTES C*n* = CAUTION Cn: measure / confirm before you connect. Refer to spec § 7.0.

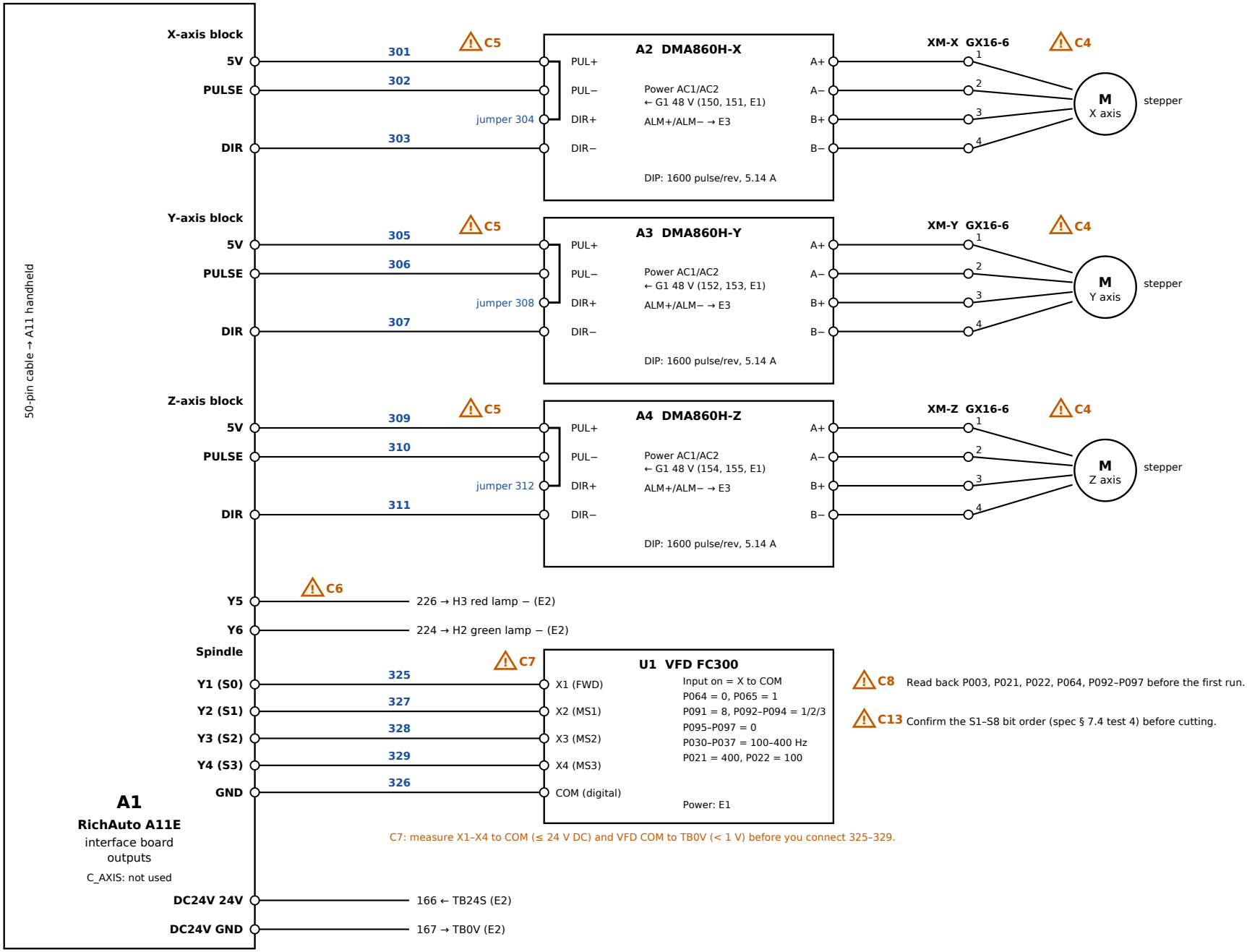
1. All inputs are low-level active. An input is active when it is connected to GND (0 V).
2. B1–B6 are NPN NO sensors. Their outputs can connect in parallel. X+, Y+ and Z− connect to X6 through TB-HL.
3. Set levels in MACHINE SETUP → Voltage Setup: X1–X4, X6 = NO; X7 = NC. X8 is not connected (disable in Input Confi).
4. X5: do the ALM polarity test (spec § 7.4 test 3) before you connect. If the opto conducts in normal operation, connect the three ALM outputs in series and set X5 to NC.
5. X5–X7 functions are fixed (A11 manual). Measure sensor supply polarity before you connect (C10). Do not connect Shield.

CNC 3-Axis Router — United Control Box (RichAuto A11)

RichAuto A11 inputs

Sheet E3 of E4 · Rev 0.5 · 2026-10-19

IEC 60617 symbols · Text: ASD-STE100 · Wire nos. = spec § 8



C8 Read back P003, P021, P022, P064, P092–P097 before the first run.

C13 Confirm the S1–S8 bit order (spec § 7.4 test 4) before cutting.

NOTES C*n* = CAUTION Cn: measure / confirm before you connect. Refer to spec § 7.0.

1. Axis outputs: common anode, 8 mA max (A11). DMA860H needs 7–16 mA (10 typ.). Jumper PUL+ to DIR+ (304, 308, 312).
2. Use twisted wires for each axis. Do not connect SHIELD (one box, less than 1 m).
3. DMA860H DIP: 1600 pulse/rev, 5.14 A peak. Set the A11 pulse equivalent to 320 pulse/mm (5 mm pitch).
4. Y1–Y4 connect directly to VFD X1–X4 (FC300: input on = X connected to COM). No relays are necessary.
5. The board has no 0–10 V output. Spindle speed = multi-speed, 8 levels (spec § 7.3 speed table).
6. Motor cores A+/A−/B+/B− → GX16-6 pins 1–4. Do not connect or disconnect motors when QF2 is ON (C4).

CNC 3-Axis Router — United Control Box (RichAuto A11)

RichAuto A11 outputs — axis drives and spindle

Sheet E4 of E4 · Rev 0.5 · 2026-10-19

IEC 60617 symbols · Text: ASD-STE100 · Wire nos. = spec § 8